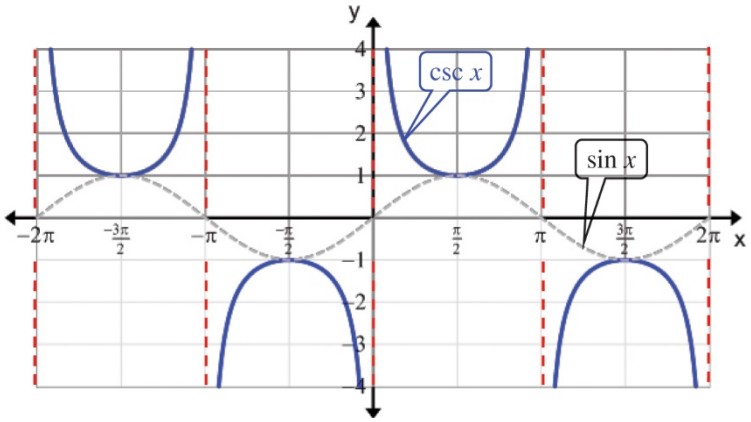
Graph the Secant and Cosecant Functions

To graph variations of $y = \csc x$ or $y = \sec x$, we use the graph of the related reciprocal function for reference.

$y = \csc x = \dfrac{1}{\sin x}$ If $\sin x = 0$, then $\dfrac{1}{\sin x}$ is undefined, and the graph of $y = \csc x$ has a vertical asymptote.

This occurs when x is a multiple of π: $x = n\pi$ for all integers n.

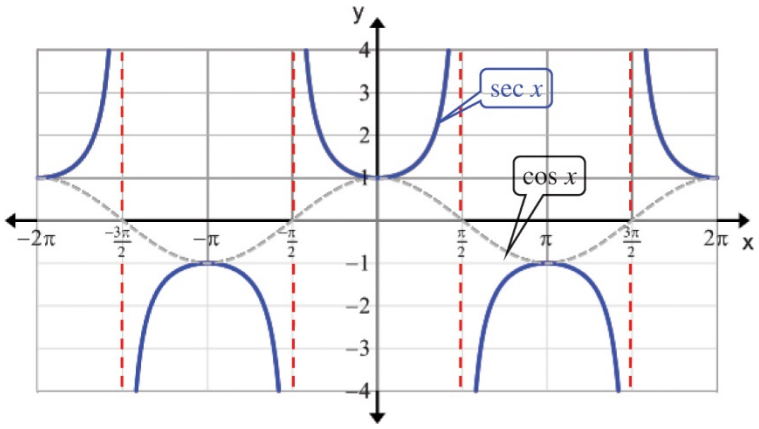


The relative maxima and minima from the sine function correspond to the relative minima and maxima for the cosecant function. Similarly with the secant function below.

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
sin x	0	1	0	−1	0
csc x	Und.	1	Und.	−1	Und.

$y = \sec x = \dfrac{1}{\cos x}$ If $\cos x = 0$, then $\dfrac{1}{\cos x}$ is undefined, and the graph of $y = \sec x$ has a vertical asymptote.

This occurs when x is an odd multiple of $\dfrac{\pi}{2}$: $x = \dfrac{(2n+1)\pi}{2}$ for all integers n.



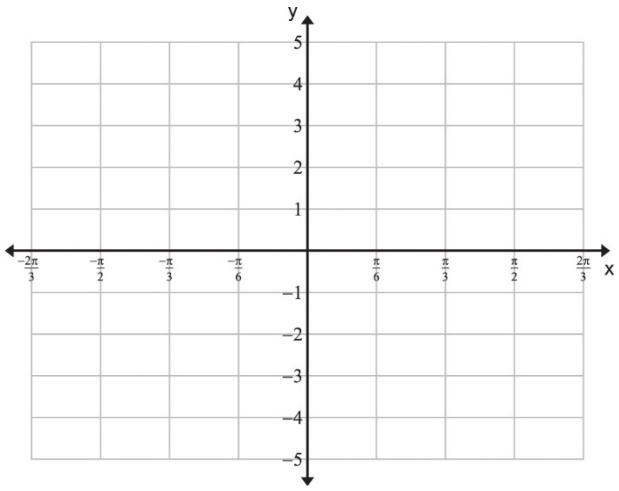
x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
cos x	1	0	−1	0	1
sec x	1	Und	−1	Und.	1

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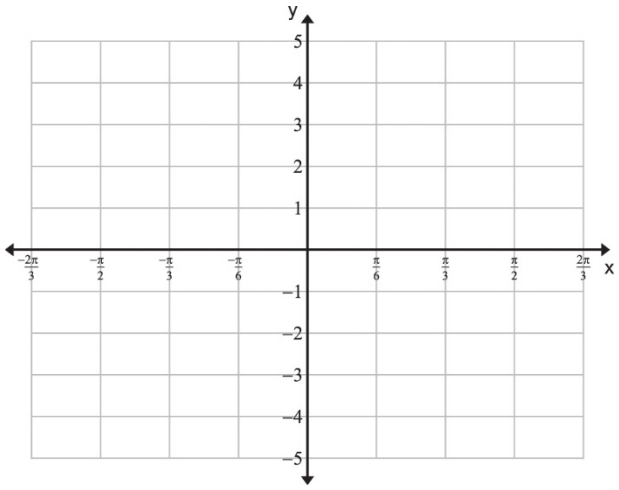
Graphs of the Cosecant and Secant Functions		
	$y = \csc x$	$y = \sec x$
Domain	$\{x \mid x \neq n\pi \text{ for all integers } n\}$	$\left\{x \mid x \neq \frac{(2n+1)\pi}{2} \text{ for all integers } n\right\}$
Range	$\{y \mid y \leq -1 \text{ or } y \geq 1\}$	$\{y \mid y \leq -1 \text{ or } y \geq 1\}$
Amplitude	None ($y = \csc x$ increases and decreases without bound)	None ($y = \sec x$ increases and decreases without bound)
Period	2π	2π
Vertical Asymptotes	$x = n\pi$ (multiples of π)	$x = \frac{(2n+1)\pi}{2}$ (odd multiples of $\frac{\pi}{2}$)
Symmetry	Origin (csc is an odd function.)	y-axis (sec is an even function.)

1.

a. Graph $y = -2\csc 3x$



b. Graph $y = -2\csc 3x + 1$



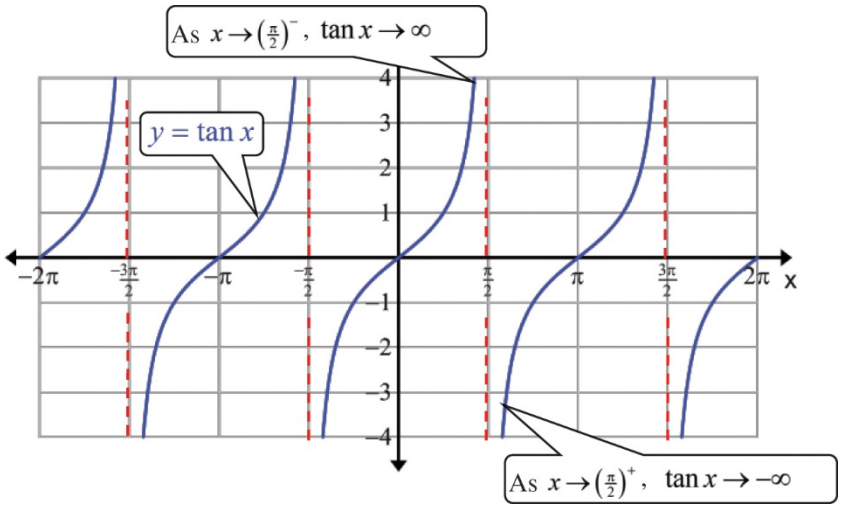
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Graph the Tangent and Cotangent Functions

$y = \tan x = \dfrac{\sin x}{\cos x}$

x-intercept where $\sin x = 0$; x is a multiple of π: at $x = n\pi$ for integer n.

vertical asymptote where $\cos x = 0$; x is an odd multiple of $\dfrac{\pi}{2}$: at $x = \dfrac{(2n+1)\pi}{2}$ for integer n.



The tangent function is an odd function and is symmetric with respect to the origin.

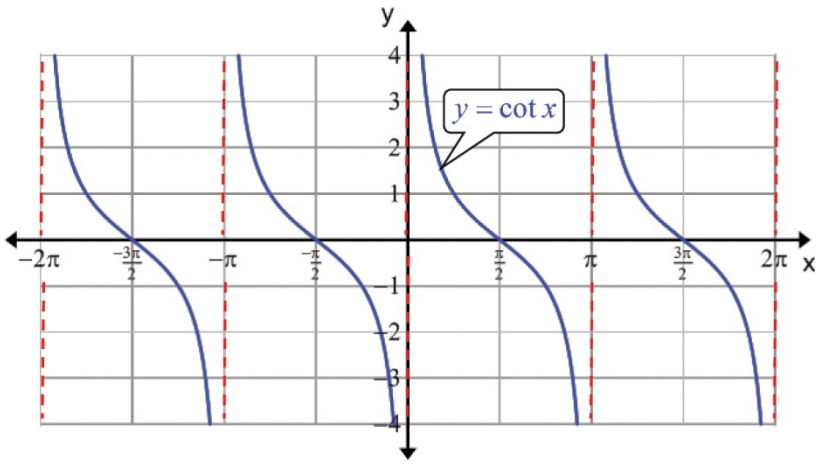
Each Quadrant I point has a mirror image in Quadrant III.

The period of the tangent function is π, so one complete period occurs between two consecutive vertical asymptotes.

$y = \cot x = \dfrac{\cos x}{\sin x}$

x-intercept where $\cos x = 0$; x is an odd multiple of $\dfrac{\pi}{2}$: at $x = \dfrac{(2n+1)\pi}{2}$ for integer n.

vertical asymptote where $\sin x = 0$; x is a multiple of π: at $x = n\pi$ for integer n.



The cotangent function is an odd function and is symmetric with respect to the origin.

Each Quadrant I point has a mirror image in Quadrant III.

The period of the cotangent function is π, so one complete period occurs between two consecutive vertical asymptotes.

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Graphs of the Tangent and Cotangent Functions		
	$y = \tan x$	$y = \cot x$
Domain	$\left\{x \mid x \neq \frac{(2n+1)\pi}{2} \text{ for all integers } n\right\}$	$\{x \mid x \neq n\pi \text{ for all integers } n\}$
Range	All real numbers	All real numbers
Amplitude	None ($y = \tan x$ is unbounded.)	None ($y = \cot x$ is unbounded.)
Period	π	π
Vertical Asymptotes	$x = \frac{(2n+1)\pi}{2}$ (odd multiples of $\frac{\pi}{2}$)	$x = n\pi$ (multiples of π)
Symmetry	Origin (tan is an odd function.)	Origin (cot is an odd function.)

Graphing $y = A\tan(Bx - C) + D$ and $y = A\cot(Bx - C) + D$ with $B > 0$. (If $B < 0$, rewrite the equations.)

The period is $\frac{\pi}{B}$. The phase shift is $\frac{C}{B}$. The vertical shift is D.

$|A|$ is the vertical scaling factor. If $A < 0$, the graph is reflected across the x-axis.

1. Find an interval between two consecutive vertical asymptotes and graph the asymptotes as dashed lines.

   $y = A\tan(Bx - C)$ and $B > 0$, solve the inequality $-\frac{\pi}{2} < Bx - C < \frac{\pi}{2}$.

   $y = A\cot(Bx - C)$ and $B > 0$, solve the inequality $0 < Bx - C < \pi$.

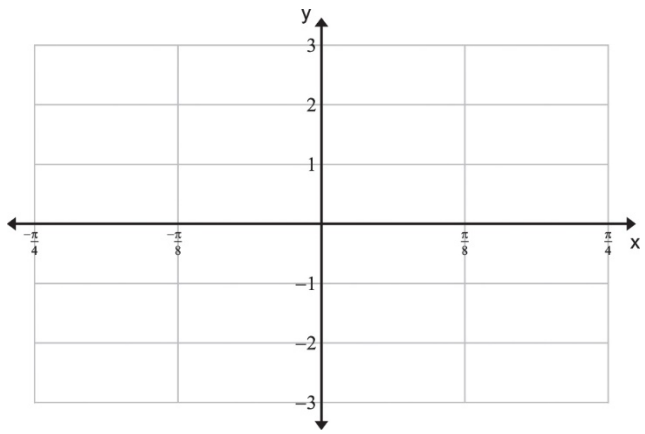
2. Plot an x-intercept halfway between the asymptotes from Step 1.

3. Sketch the general shape of the "parent" function between the asymptotes.
   If more accuracy is desired, consider evaluating the function for x values midway between the x-intercept and the asymptotes found in step 1.

   For $y = \tan x$, the y-coordinates will be $-A$ and A.
   For $y = \cot x$, the y-coordinates will be A and $-A$.

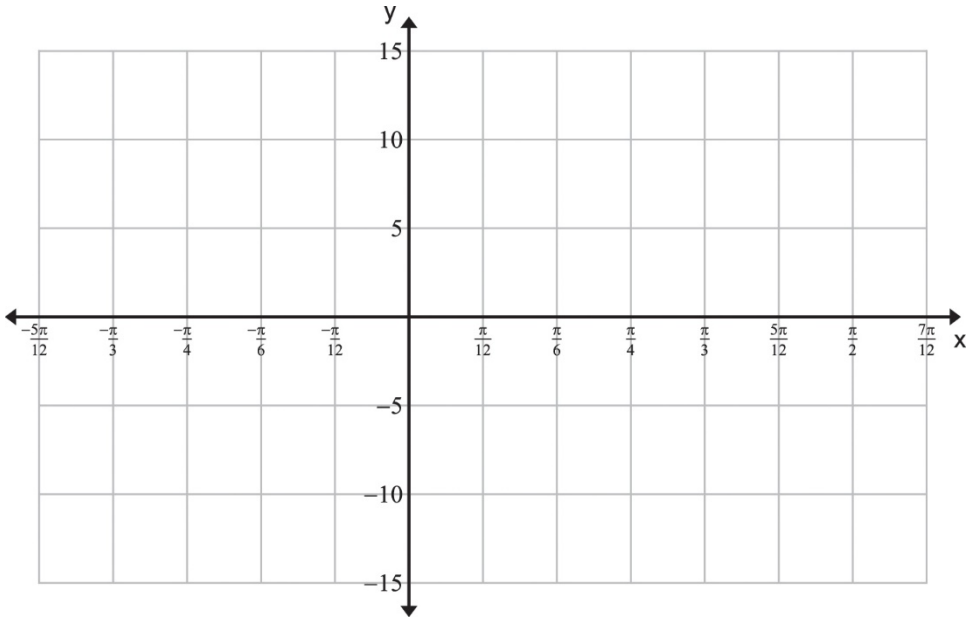
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2. Graph $y = \cot 4x$

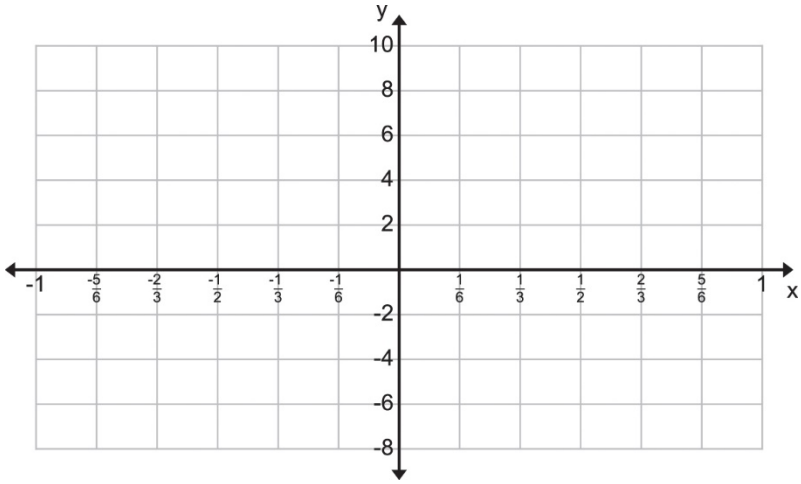


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3. Graph $y = 4\tan\left(2x + \frac{\pi}{3}\right)$



4. Graph $y = -\tan\frac{3\pi}{4}x + 2$



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